LPCUNet:A Lightweight Pure CNN UNet for Efficient Urban Scene Remote Sensing Semantic Segmentation

*Bing Liu
College of Computer Science and Engineering
Changchun University of Technology
College of Computer Science and Technology
Jilin University
Changchun, China
*liubing@ccut.edu.cn

Huizhu Wu
College of Computer Science and Engineering
Changchun University of Technology
Changchun, China
626057532@qq.com

Xueliang Bao
Aviation University of Air Force
Changchun, China
15947875222@163.com

Zhaohao Zhong
College of Computer Science and Engineering
Changchun University of Technology
Changchun, China
2202103113@stu.ccut.edu.cn

2023 4th International Conference on Computer Vision, Image and Deep Learning (CVIDL) | 979-8-3503-2644-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/CVIDL58838.2023.10167334

***Abstract*—Semantic segmentation, a fundamental task in computer vision, has developed rapidly in recent years. Semantic segmentation of remote sensing urban scene images, utilized in tasks such as land cover mapping, urban change detection, environmental preservation, and economic assessment, has also received much attention. In order to extract global semantic features, recent research has focused on combining transformers with CNN, which have great potential in global information modeling, for semantic segmentation models. However, the hybrid method of CNN and Vision Transformer still suffers from low latency. And the MLP of the visual transformer results in many parameters. To address these issues, we propose a lightweight pure CNN UNet(LPCUNet) model, which introduces a sizeable convolutional kernel to capture the global context. Moreover, a simple fusion module dynamically fuses local and global features. Extensive experiments show that our proposed method achieves state-of-the-art performance while faster latency. More precisely, the LPCUNet model demonstrated impressive performance with 83.3% and 86.4% mean Intersection over Union (mIoU) scores on the Vaihingen and Potsdam datasets respectively.**

Keywords-remote sensing; lightweight; CNN; semantic segmentation

I. INTRODUCTION

Semantic segmentation aims to predict a label for each pixel in an image. It has gained extensive utilization as a fundamental aspect of computer vision, finding applications in various domains including but not limited to autonomous driving, medical image analysis, 3D reconstruction, and numerous others. Remote sensing satellites provide a large amount of data, and remote sensing images are becoming increasingly important in many applications, including land use monitoring [1], precision agriculture [2], disaster prevention [3], wildfire detection [4], vector-borne disease surveillance [5], and climate change response [6]. In view of the rich spatial complexity and important Semantic information in remote sensing images, semantic segmentation of urban scenes from remote sensing data is essential for many practical purposes. These include land cover mapping, urban change detection, environmental protection and economic assessment, and accurate delineation of urban areas. Due to the high spatial scale of remote sensing images. A standard method is to design a lightweight CNN and Vision Transformer hybrid model to extract the global semantic feature with local detail information(e.g., SwinUNet [7], TransUNet [8]). However, the CNN and Vision Transformer hybrid methods still suffer from lower latency. Furthermore, the MLP of the vision transformer results in many parameters[18].

To solve this problem, we propose a lightweight pure CNN UNet (LPCUNet) model, which is a pure CNN model based on the Unet architecture. In order to extract the global semantic feature efficiently. We introduce the big kernel size convolution or not the vision transformer module to extract global context features. Furthermore, we propose a simple feature fused (SF) module, which introduces a dynamic mechanism to fuse local and global features. Numerous experiments show that our method can achieve state-of-the-art performance.

Overall, our contributions can be outlined as follows:

- We propose a lightweight pure CNN model, LPCUNet, to fix the low latency of vision transformer.
- We propose a CNN-based decoder that can extract global features and fuse features by big kernel convolution.
- Our method achieves state-of-the-art performance.

II. METHOD

This section provides a detailed introduction to the proposed lightweight pure convolution (LPCUNet) model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:56:55 UTC from IEEE Xplore. Restrictions apply.

Firstly, we describe the overall structure of LPCUNet. Then, we elaborated on the proposed encoder and decoder. Finally, we introduce the loss function for monitoring the proposed network.

Figure 1 illustrates the overall architecture, which comprises an encoder and a decoder component. We define the Resnet18 [9] as an encoder; the rest of the model is the decoder. For the encoder, we introduce the most common pre-trained ResNet18 to extract local features. For the decoder, to reduce computational costs and better extract global semantic features, we adopt a 7×7 convolution to extract global semantic features. Furthermore, the decoder also contains SF module, which introduces dynamic mechanisms to fuse local and global features.

A. CNN-based encoder

The encoder is a four-stage backbone. For input image $I \in \mathbb{R}^{\mathrm{H}\times W\times 3}$, we input it into the encoder and output a feature map $x_o \in \mathbb{R}^{\frac{H}{32}\times\frac{W}{32}\times C}$ for the decoder input. We also need the local feature $x_{eo}^{k} \in \mathbb{R}^{\frac{H}{2^{k+1}}\times\frac{W}{2^{k+1}}\times C}, k \in \{1,2,3\}$ to inject detailed information into the global semantic feature.

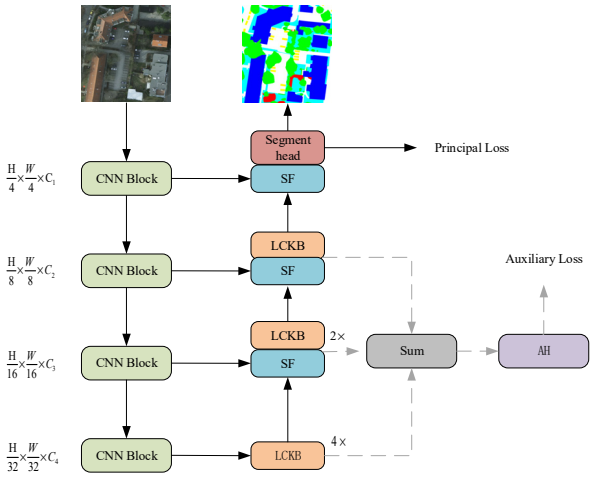


Fig. 1.Framework of LPCUNet.

B. Large Convolutional Kernel-based decoder :

In order to extract global semantic features during the decoding stage, a common method is to introduce transformer to extract global semantic features, but the MLP of the vision transformer resulted in many parameters. On the contrary, in our proposed LPCUNet, to reduce computational complexity. We construct a low-cost decoder to utilize three 7×7 convolution and batch norm layers and one feature refinement head. The decoder also includes a simple feature fusion (SF) module that injects the local detail information through the dynamic mechanism.

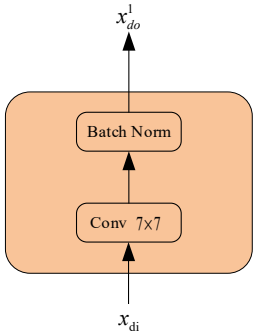


Fig. 2.The structure of LCKB.

First, we introduce a 1×1 convolutional kernel on the $x_{di} \in \mathbb{R}^{H'\times W'\times C}$ to get the input of the decoder. Then we feed x_{di} to the decoder, as illustrated in **Fig. 2**. The feature x_{di} first through a LCKB block to get the global semantic feature x_{do}^{1}.

Then we feed $x_{eo}^{i}, i \in \{1,2,3\}$ and $x_{do}^{i}, i \in \{1,2,3\}$ into the SF module to calculate the fusion feature SF using formula (6). We can get the coarse global semantic feature map thrice by the above. Then we introduce a simple head to fine-grain the coarse feature map for the final prediction. Formally, the process of the decoder can be written as follows:

$$x_{do}^{1} = LCKB_1(x_{di}) \tag{1}$$

$$x_{do}^{2} = LCKB_2(SF(x_{do}^{1}, x_{eo}^{1})) \tag{2}$$

$$x_{do}^{3} = LCBK_3(SF(x_{do}^{2}, x_{eo}^{2})) \tag{3}$$

$$x_{do} = FRH(SF(x_{do}^{3}, x_{eo}^{3})) \tag{4}$$

Where the LCKB is the convolutional blocks with large convolutional kernels. x_{di} is the input of the decoder. $x_{eo}^{i}, i \in \{1,2,3\}$ denotes the feature generated by the encoder block. $x_{do}^{i}, i \in \{1,2,3\}$ signifies the feature outputted by the LCBK, SF is the simple feature fusion module.

The LCKB consists of a 7×7 convolutional and a batch norm layer. The process of LCKB can be written as follows:

$$LCKB(\cdot) = Batchnorm(Conv_{7\times7}(\cdot)) \tag{5}$$

where $\mathrm{Conv}_{7\times7}(\cdot)$ means 7×7 means convolution and $Batchnorm(\cdot)$ means batch norm layer.

The simple feature fusion (SF) module fuses local and global features through dynamic mechanisms. as shown in **Fig. 3**. The SF module injects detailed information by dynamic fuse the local feature and global semantic feature. First, the SF module seems x_{eo}^{i} and to x_{do}^{i} as input. Then we add two learnable parameters and act them to get the dynamic weight. Moreover, we dot the dynamic weight and local and global features to inject detailed information. It can be written as:

$$SF = x_{eo}^{i} \cdot \sigma\left(ConvBN(x_{eo}^{i} + b_1)\right) + x_{do}^{i} \cdot \sigma\left(ConvBN(x_{eo}^{i} + b_2)\right), i \in \{1,2,3\} \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:56:55 UTC from IEEE Xplore. Restrictions apply.

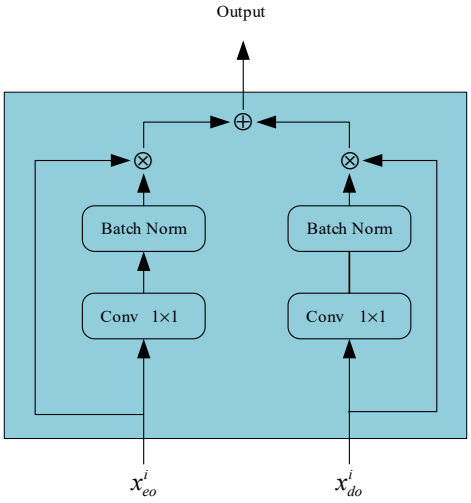


Fig. 3.The structure of SF.

Where SF represents the fused feature, x_{eo}^{i} denotes the feature produced by the encoder block, and x_{do}^{i} indicates the feature generated by the decoder block(LCBK), σ is the RELU6, $ConvBN(\cdot)$ indicates a operation with convolutional and Batch Normalization.

Finally, we select the cross-entropy loss as the loss function, and we use an auxiliary loss to accelerate training speed and improve training effectiveness.

III. EXPERIMENTS

A. Experimental setup

Datasets:

Vaihingen: Composed of 33 very fine spatial resolution TOP images, with an average pixel size of 2494×2064 pixels. The dataset consists of five foreground categories (impermeable surfaces, buildings, low vegetation, trees, cars) and one background category (clutter).

Potsdam: Contains 38 highly refined spatial resolution TOP image tiles, with a size of 6000×6000 pixels and the same category information as Vaihingen.

Implementation details:

All experimental models were implemented using the PyTorch framework and trained on an NVIDIA Tesla P40 GPU. To ensure rapid convergence, the AdamW optimizer was utilized for training all models. A base learning rate of 6e-4 was set, and a cosine learning rate schedule was employed to adjust the learning rate during training.

For the Potsdam dataset, we selected a training set consisting of 23 images and the other 14 images to test. For the Vaihingen dataset, we chose 17 images for testing and the remaining 16 images for training. The images of the above datasets were randomly cropped into 512×512 patches. We adopted enhancement techniques during the training process, such as random horizontal flipping, random rotation, and random scale random vertical flipping, and, with training epoch set to 100 and batch size of 16. We applied multi-scale and random flip enhancement techniques during the testing phase.

Evaluation metrics:

In the experiment, we introduce two main types of evaluation indicators, including mean union intersection (mIoU) and inference.

B. Experiment results:

According to the results presented in Tables 1 and 2, LPCUNet provides the best mIoU significantly superior to CNN and Transformer based lightweight networks on the Vaihingen and Potsdam test sets. Additionally, **Fig. 4** show cases the visualization results, demonstrating the accurate segmentation of small objects and boundaries achieved by our method.

TABLE 1.THE COMPARISON RESULTS OF VAIHINGEN.

Method	Backbone	Imp.surf	Building	Lowveg.	Tree	Car	Inference(s)	mIoU
MAResU-Net [10]	ResNet18	92.0	95.0	83.7	89.3	78.3	0.69	78.6
SwiftNet [10]	ResNet18	92.2	94.8	84.1	89.3	81.2	3.46	79.6
ABCNet [12]	ResNet18	92.7	95.2	84.5	89.7	85.3	4.68	81.3
BoTNet [13]	ResNet18	89.9	92.1	81.8	88.7	71.3	3.55	74.3
Segmenter [15]	ViT-Tiny	89.8	93.0	81.2	88.9	67.6	32	73.6
LPCUNet	ResNet18	93.0	95.5	85.1	90.7	89.4	0.62	83.3

TABLE 2.THE COMPARISON RESULTS OF POSTDAM.

Method	Backbone	Imp.surf	Building	Lowveg.	Tree	Car	mIoU
MAResUNet [10]	ResNet18	91.4	95.6	85.8	86.6	93.3	83.9
SwiftNet [10]	ResNet18	91.8	95.9	85.7	86.8	94.5	83.8
BANet [14]	ResT-Lite	93.3	96.7	87.4	89.1	96.0	86.3
EaNet [16]	ResNet18	92.0	95.7	84.3	85.7	95.1	83.2
FANet [17]	ResNet18	92.0	96.1	86.0	87.8	94.5	84.2
LPCUNet	ResNet18	93.3	96.9	87.6	88.5	96.3	86.4

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:56:55 UTC from IEEE Xplore. Restrictions apply.

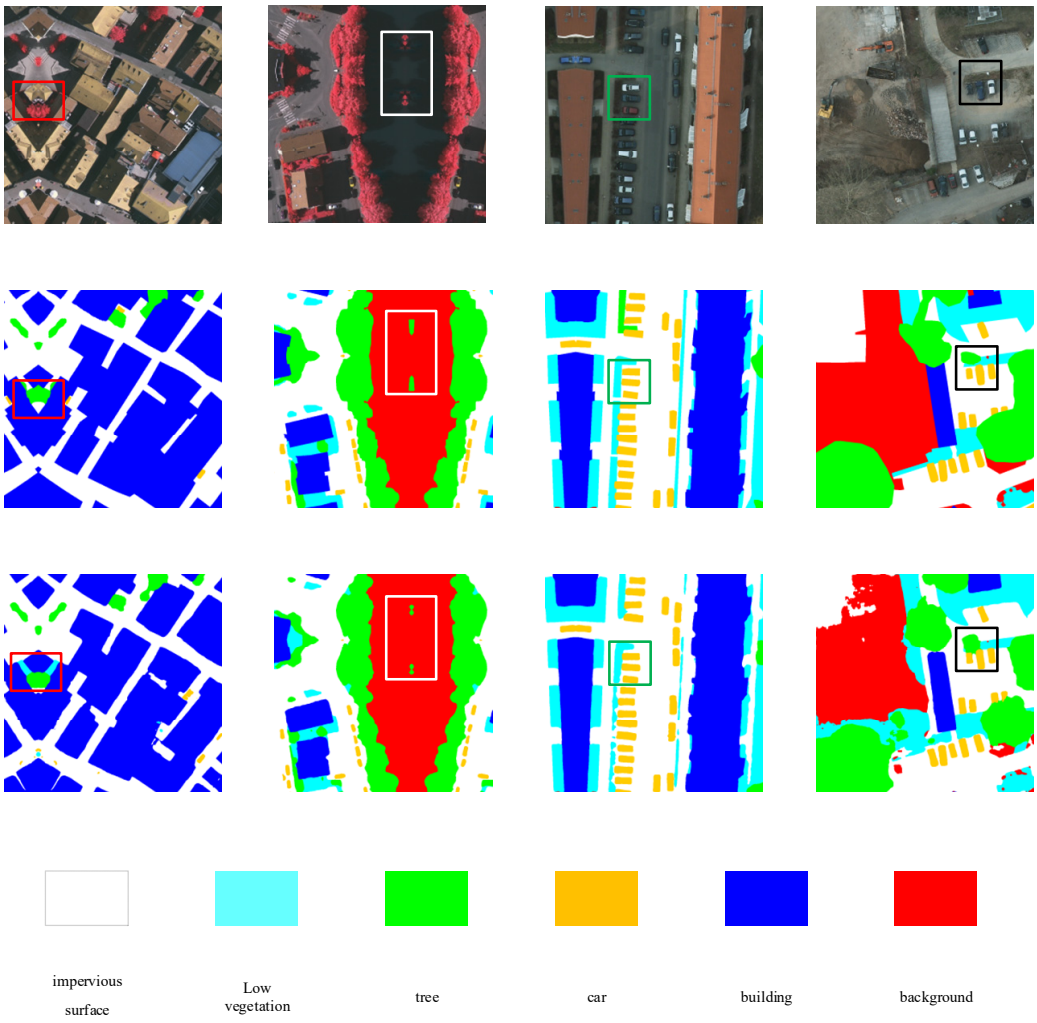


Fig. 4.Visualization results of the Vaihingen and Potsdam. The upper row denotes the input RGB images. The middle row represents the ground truth. The bottom row is our proposed LPCUNet. The first and second columns are from the Vaihingen test set, and the rest columns are from the Potsdam test set.

C. *Ablation study*

To individually assess the performance of each component in the LPCUNet model, we conducted a series of ablation experiments. The outcomes are presented and detailed in TABLE 3 and TABLE 4.

TABLE 3.ABLATION STUDY OF EACH COMPONENT OF THE LPCUNET.

Dataset	Method	mIoU(%)
Vaihingen	Baseline	77.1
	Baseline+LCKB	81.8
	Baseline+LCKB+SH	83.3

TABLE 4.ABLATION STUDY WITH DIFFERENT CONVOLUTIONAL KERNEL SIZES OF THE LPCUNET.

Dataset	7×7	9×9	11×11	mIoU(%)
Vaihingen	√			83.3
		√		83.3
			√	83.2

IV. CONCLUSIONS

In this article, we propose a pure CNN decoder with a large convolutional kernel and construct a UNet-like model for effective semantic segmentation of remote sensing urban scene images. In order to alleviate the computational burden brought by transformer-based decoders, we choose large convolutional kernels instead of the vision transformer to encode global features. In addition, we propose a simple feature fusion (SF) module that introduces dynamic mechanisms to fuse local and global features. The complete experimental results show that the proposed method achieves state-of-the-art segmentation accuracy.

References

[1] Foody G M. Remote sensing of tropical forest environments: towards the monitoring of environmental resources for sustainable development[J]. International journal of remote sensing, 2003, 24(20): 4035-4046.

[2] Mulla D J. Twenty five years of remote sensing in precision agriculture: Key advances and remaining knowledge gaps[J]. Biosystems engineering, 2013, 114(4): 358-371.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:56:55 UTC from IEEE Xplore. Restrictions apply.

[3] Schumann G J P, Brakenridge G R, Kettner A J, et al. Assisting flood disaster response with earth observation data and products: A critical assessment[J]. Remote Sensing, 2018, 10(8): 1230.

[4] Filipponi F. Exploitation of sentinel-2 time series to map burned areas at the national level: A case study on the 2017 italy wildfires[J]. Remote Sensing, 2019, 11(6): 622.R. Nicole, "Title of paper with only first word capitalized," J. Name Stand. Abbrev., in press.

[5] Ippoliti C, Candeloro L, Gilbert M, et al. Defining ecological regions in Italy based on a multivariate clustering approach: A first step towards a targeted vector borne disease surveillance[J]. PloS one, 2019, 14(7): e0219072.

[6] Rolnick D, Donti P L, Kaack L H, et al. Tackling climate change with machine learning[J]. ACM Computing Surveys (CSUR), 2022, 55(2): 1-96.

[7] Cao H, Wang Y, Chen J, et al. Swin-unet: Unet-like pure transformer for medical image segmentation[C]//Computer Vision–ECCV 2022 Workshops: Tel Aviv, Israel, October 23–27, 2022, Proceedings, Part III. Cham: Springer Nature Switzerland, 2023: 205-218.

[8] Chen J, Lu Y, Yu Q, et al. Transunet: Transformers make strong encoders for medical image segmentation[J]. arXiv preprint arXiv:2102.04306, 2021.

[9] He K, Zhang X, Ren S, et al. Deep residual learning for image recognition[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 770-778.

[10] Oršić M, Šegvić S. Efficient semantic segmentation with pyramidal fusion[J]. Pattern Recognition, 2021, 110: 107611.Li R, Zheng S, Duan C, et al. Multistage attention ResU-Net for semantic segmentation of fine-resolution remote sensing images[J]. IEEE Geoscience and Remote Sensing Letters, 2021, 19: 1-5.

[11] Oršić M, Šegvić S. Efficient semantic segmentation with pyramidal fusion[J]. Pattern Recognition, 2021, 110: 107611.

[12] Li R, Zheng S, Zhang C, et al. ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of Fine-Resolution remotely sensed imagery[J]. ISPRS Journal of Photogrammetry and Remote Sensing, 2021, 181: 84-98.

[13] Srinivas A, Lin T Y, Parmar N, et al. Bottleneck transformers for visual recognition[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021: 16519-16529.

[14] Wang L, Li R, Wang D, et al. Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images[J]. Remote Sensing, 2021, 13(16): 3065.

[15] Strudel R, Garcia R, Laptev I, et al. Segmenter: Transformer for semantic segmentation[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2021: 7262-7272.

[16] Zheng X, Huan L, Xia G S, et al. Parsing very high resolution urban scene images by learning deep ConvNets with edge-aware loss[J]. ISPRS Journal of Photogrammetry and Remote Sensing, 2020, 170: 15-28.

[17] Hu P, Perazzi F, Heilbron F C, et al. Real-time semantic segmentation with fast attention[J]. IEEE Robotics and Automation Letters, 2020, 6(1): 263-270.

[18] Li, H., Huang, J., Jin, P., Song, G., Wu, Q., & Chen, J. (2022). Toward 3d spatial reasoning for human-like text-based visual question answering. arXiv preprint arXiv:2209.10326.

APPENDIX

We put all of the formula of process in the Appendix, which as follows:

$$x_{do}^{1} = LCKB_{1}(x_{di}) \tag{1}$$

$$x_{do}^{2} = LCKB_{2}(SF(x_{do}^{1}, x_{eo}^{1})) \tag{2}$$

$$x_{do}^{3} = LCBK_{3}(SF(x_{do}^{2}, x_{eo}^{2})) \tag{3}$$

Above formula is the process of extracting global semantic feature and fusing local feature and global feature to capture local-global context.

Then we introduce the FRH head to inject the local detail information. The formula an be written as follows:

$$x_{do} = FRH(SF(x_{do}^{3}, x_{eo}^{3})) \tag{4}$$

And the calculation of LCKB and SF can be written as follows:

$$LCKB(\cdot) = Batchnorm(Conv_{7\times7}(\cdot)) \tag{5}$$

$$SF = x_{eo}^{i} \cdot \sigma\left(ConvBN(x_{eo}^{i} + b_1)\right) + x_{do}^{i} \cdot \sigma\left(ConvBN(x_{eo}^{i} + b_2)\right), i \in \{1,2,3\} \tag{6}$$

The whole formulas of our proposed decoder are the above formulas. And we can get the accurate segment results by this way.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:56:55 UTC from IEEE Xplore. Restrictions apply.